

Essentials in Text and Speech Processing – Project Proposal

Robustness of Multilingual Intent Classification to Typos

Ke Sun Thi Tho Pham Tristan Hein

Research question and motivation

Intent classification is used in systems such as virtual assistants, where users often enter short and imperfect text. Our project asks how strongly realistic spelling mistakes affect multilingual intent classifiers and whether some representations or simple robustness methods handle this noise better than others.

We are especially interested in whether the effect differs across languages and model types. This provides a clear classification task while allowing us to study robustness, multilingual NLP and model behavior.

Dataset

We will use MASSIVE 1.1, a multilingual natural-language-understanding dataset with more than one million labeled utterances and 60 intent classes [1]. We will focus on English, Vietnamese, German and Simplified Chinese, using the official training, development and test splits.

Experimental approach

We will create controlled noisy versions of clean utterances by introducing character-level spelling errors such as deletions, swaps, duplications and substitutions. We plan to test multiple perturbation types or noise levels instead of relying on a single corruption setting.

We plan to compare word-level TF-IDF with Logistic Regression, character-level TF-IDF with Logistic Regression, and a fine-tuned XLM-R-base model. We will compare clean-data training with training on a mixture of clean and typo-augmented data [2], with augmentation restricted to the training split. Exact model settings and hyperparameters will be finalized after pilot experiments.

Evaluation and analysis

We will evaluate models on clean and corrupted test data using accuracy and macro-F1. A central robustness measure will be the drop in macro-F1 between clean and corrupted input.

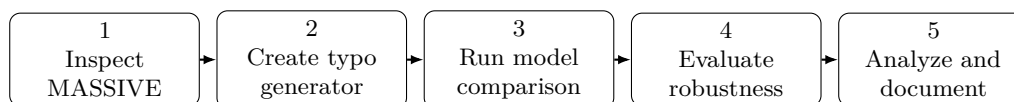
Results will be compared across languages, model types and perturbation conditions. We will also inspect representative errors or confusion matrices to understand where models fail and whether typo augmentation improves robustness without substantially reducing clean-data performance.

Value, challenges and planned output

The project aims to provide a reproducible analysis of typo robustness in multilingual intent classification. We also plan to release reusable code for generating controlled spelling noise and evaluating models.

The main challenges are creating realistic perturbations across different languages, ensuring fair model comparisons and staying within the available compute budget. We will keep the final experimental scope flexible based on early results and document the code and experiment settings for reproducibility.

Planned workflow



References

- [1] AmazonScience. *MASSIVE 1.1 dataset card*. <https://huggingface.co/datasets/AmazonScience/massive>.
- [2] S. Sengupta, J. Krone, and S. Mansour. *On the Robustness of Intent Classification and Slot Labeling in Goal-oriented Dialog Systems to Real-world Noise*. Proceedings of the 3rd Workshop on NLP for Conversational AI, 2021. <https://aclanthology.org/2021.nlp4convai-1.7/>